

KERSTINBO, Sweden

WMO: 024820

Lat: 60.2683N

Lon: 16.9738E

Elev: 57

StdP: 100.64

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.6	(c) -17.2	(d) -23.0	(e) 0.5	(f) -20.5	(g) -19.4	(h) 0.7	(i) -17.0	(j) 8.8	(k) 4.8	(l) 7.6	(m) 3.7	(n) 0.6	(o) 210	(p) 0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.6	(c) 26.8	(d) 18.5	(e) 24.9	(f) 17.5	(g) 23.0	(h) 16.5	(i) 19.7	(j) 24.3	(k) 18.6	(l) 23.3	(m) 17.6	(n) 21.7	(o) 2.8	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.1	(b) 13.1	(c) 21.4	(d) 16.9	(e) 12.1	(f) 20.0	(g) 15.8	(h) 11.3	(i) 19.3	(j) 56.3	(k) 24.1	(l) 52.8	(m) 23.4	(n) 49.6	(o) 21.7	(p) 23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	6.1	5.3	-24.4	28.4	3.9	1.5	-27.2	29.4	-29.4	30.3	-31.6	31.1	-34.5	32.2
			-24.2	21.0	3.8	1.3	-26.9	21.9	-29.2	22.6	-31.3	23.4	-34.1	24.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-4.3	-3.7	-0.6	4.5	9.7	14.1	16.6	15.4	10.8	5.4	1.2	-2.3
	DBStd	8.39	5.85	5.19	4.84	3.47	3.55	3.15	2.74	2.75	3.21	3.87	4.04	5.75
	HDD10.0	2200	442	384	328	169	49	4	0	1	29	149	263	381
	HDD18.3	4667	701	617	586	416	268	133	69	98	227	401	513	639
	CDD10.0	600	0	0	0	3	40	126	204	167	52	7	0	0
	CDD18.3	26	0	0	0	0	0	4	16	6	0	0	0	0
	CDH23.3	347	0	0	0	0	11	81	191	64	1	0	0	0
	CDH26.7	54	0	0	0	0	1	9	36	9	0	0	0	0
Wind	WSAvg	2.4	2.4	2.3	2.4	2.4	2.5	2.4	2.3	2.2	2.3	2.4	2.5	2.5
Precipitation	PrecAvg	603	38	26	32	34	36	70	69	77	58	60	56	49
	PrecMax	723	70	54	67	79	75	150	132	173	123	151	118	117
	PrecMin	520	3	0	6	7	10	11	21	34	7	20	23	13
	PrecStd	58	18	14	16	21	16	31	33	32	31	29	26	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.7	14.5	19.8	24.8	28.0	30.0	27.7	22.4	16.1	11.1	10.1
		MCWB	5.1	4.5	7.9	11.6	15.0	17.5	19.4	19.2	16.1	12.8	9.8	8.6
	2%	DB	5.2	5.8	10.6	16.7	22.2	25.8	27.4	25.0	19.9	14.1	9.5	7.5
		MCWB	3.9	3.6	5.9	10.0	13.8	16.9	19.0	18.5	15.2	11.9	8.4	6.1
	5%	DB	4.1	4.3	8.1	13.9	19.4	23.5	25.6	23.0	18.0	12.7	8.1	5.8
		MCWB	3.2	2.8	4.5	8.5	12.8	15.6	18.3	17.2	14.4	11.0	7.3	4.6
	10%	DB	2.7	3.0	5.9	11.5	17.2	21.3	23.4	21.2	16.4	11.4	7.0	4.4
		MCWB	1.9	1.8	3.2	6.8	11.4	14.8	17.5	16.5	13.4	10.0	6.3	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	5.6	8.1	12.5	16.5	19.0	21.3	21.2	17.4	13.6	10.1	8.8
		MCDB	6.6	6.8	13.2	18.3	22.5	25.4	25.5	25.0	20.3	15.3	10.8	10.0
	2%	WB	4.1	4.1	6.4	10.4	14.8	17.7	20.1	19.4	15.7	12.4	8.6	6.4
		MCDB	4.9	5.3	9.9	15.7	20.0	23.7	24.9	22.8	18.9	13.6	9.3	7.2
	5%	WB	3.1	3.0	4.9	8.8	13.5	16.6	19.1	18.3	14.8	11.3	7.4	4.8
		MCDB	3.9	4.2	7.6	13.3	18.5	22.1	24.4	21.8	17.4	12.5	8.0	5.5
	10%	WB	2.0	1.8	3.6	7.3	12.2	15.4	18.0	17.2	13.8	10.1	6.4	3.5
		MCDB	2.7	2.8	5.5	10.6	16.5	20.2	22.7	20.4	15.9	11.1	7.0	4.2
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.9	8.7	10.3	11.0	10.9	10.6	10.0	9.3	7.2	4.8	5.3
		MCDBR	5.4	7.0	11.6	15.2	15.9	15.5	14.9	13.4	11.8	8.2	6.1	5.6
		MCWBR	5.0	5.3	7.6	9.1	8.9	7.9	7.4	7.5	7.6	6.1	5.5	5.0
	5% WB	MCDBR	5.3	6.0	10.6	13.7	13.1	13.7	12.4	10.9	10.6	7.1	5.7	5.4
		MCWBR	5.0	4.8	7.2	8.4	7.8	7.5	6.6	6.7	7.1	5.7	5.2	5.2
Clear-Sky Solar Irradiance	taub		0.264	0.309	0.331	0.365	0.364	0.356	0.379	0.373	0.338	0.325	0.289	N/A
	taud		2.196	2.283	2.362	2.347	2.409	2.458	2.412	2.425	2.511	2.439	2.251	N/A
	Ebn at Noon		507	686	799	832	863	876	842	818	792	670	467	N/A
	Edh at Noon		40	66	84	102	104	100	103	94	73	57	38	N/A
All-Sky Solar Radiation	RadAvg		0.28	0.92	2.41	3.87	4.97	5.52	5.11	4.01	2.62	1.18	0.36	0.18
	RadStd		0.04	0.15	0.29	0.38	0.50	0.39	0.51	0.41	0.32	0.19	0.09	0.02

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-110	N/A	N/A	N/A

Nomenclature: See separate page